

Owen Wu

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

September 2026 – Present

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Frameworks: Next.js, React, Django, Flask, Tailwind CSS, discord.js

Backend: RESTful APIs, MongoDB, Mongoose, Google Cloud Platform, Redis

Developer Tools: Git, GitHub, Jira, Docker, Vercel, Upstash, Power BI, Power Apps, Power Automate, L^AT_EX

EXPERIENCE

Software Engineering Intern (Future Tech)

July 2025 – August 2025

Nokia

- Developed a centralized document catalog using Power Apps, managing 600+ reports for the Data Analytics team
- Automated reporting workflows using Power Automate and Excel APIs, eliminating manual stakeholder alerts
- Optimized Power BI dashboards with a 10-member analytics team, improving report performance and usability

Full-stack Developer

August 2025 – June 2026

JAMHacks

- Implemented Sponsorship and Teams styling using Next.js and Tailwind CSS, serving 10k+ website visitors
- Built REST APIs using Next.js and MongoDB, automating applicant acceptance, waitlist, and rejection workflows
- Developed Discord bots using Discord.js, MongoDB, and GCP, assisting 250+ hackers with mentor ticket handling
- Integrated GCP Vertex AI with MongoDB, summarizing 250+ participant data for a 19-member organizing team

PROJECTS

StarWeave

May 2026 - Present

- Developed a full-stack astronomy database using Next.JS, TypeScript, MongoDB, and Redis with 7k+ objects
- Built REST APIs that reduced database retrieval time from 3.0s to 1.2s through optimized MongoDB queries
- Designed a Redis/MongoDB caching layer, reducing API response and page load times by 87% (1.5s → 0.3s)
- Secured REST APIs using Upstash rate limiting, protecting backend services from abuse and excessive requests
- Integrated data from NASA JPL Horizons API and Exoplanet API into a scalable backend data pipeline

DMOJ Open-Source Contributor

May 2026 - Present

- Implemented dynamic username styling using Django and CSS to reflect rank-based username colour assignments
- Developed a immutable identifier system using Python, Django, and SQL that allows site-wide username changes

Graph Theory Algorithm Visualizer

December 2025 - January 2026

- Built an interactive graph theory visualizer using Java and Processing for custom mathematical graph creation
- Implemented BFS and DFS visualizations using OOP, animating graph traversals with explanations in real time

AWARDS AND ACHIEVEMENTS

Canadian Computing Contest Senior: Used Python, Java, and C++ to solve programming problems that cover topics from mathematics to various data structures and algorithms. Top 25% of participants in 2024, 2025, and 2026

Best Use of MATLAB, JAMHacks 8: Developed a MATLAB back-end to perform real-time audio analysis with a Python Flask front-end using web sockets in our project "AnyJams!"